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Category: ALGORITHMS | Speedup: Exponential | Concept ID #47

Foundational Research & Advances in Quantum Fourier Transform (QFT)

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1. ABSTRACT & THEORETICAL FOUNDATION

This peer-reviewed reference document synthesizes the formal mathematical formulation, operational circuit protocols, and physical implementation benchmarks for Quantum Fourier Transform (QFT) in modern quantum information architectures.

2. CORE CONCEPTUAL INTUITION (PLAIN ENGLISH)

"Like tuning a radio to find the hidden musical pitch: it converts a pattern of quantum numbers into its underlying frequencies in a fraction of a second."

3. Mathematical Formulation & Unitary Rule

```
\text{QFT}|j_1 j_2 \dots j_n\rangle = \frac{1}{2^{n/2}} \bigotimes_{l=1}^n \left( |0\rangle +
```

Binary fraction expansion allows product-state representation implemented with $O(n^2)$ Hadamard and Controlled-R_k gates.

4. Step-by-Step Circuit Sequence

Alternating cascade of Hadamard gates and Controlled-Phase rotations R_2, R_3... followed by SWAP gate reversal.

5. Executable IBM Qiskit (Python) Code

```python
from qiskit import QuantumCircuit
import numpy as np

qc = QuantumCircuit(3)
# Qubit 0
qc.h(0)
qc.cp(np.pi / 2, 1, 0)
qc.cp(np.pi / 4, 2, 0)
# Qubit 1
```

6. Computational Complexity & Speedup

Classical Time:	Quantum Time:	Speedup Class:
O(n 2^n) via classical Fast Fourier Transform (FFT)	O(n^2) quantum gate complexity (or O(n log n) with bounded precision tru	Exponential

7. Key Applications & Future Research Directions

- Subroutine for Quantum Phase Estimation (QPE)
- Shor's integer factorization and discrete logarithms
- HHL algorithm for quantum linear systems

Future Work:

Hardware fidelity improvements, compiler transpilation optimizations, and integration with fault-tolerant Algorithms pipelines.